# Your Voider

Voider connects supported Ethernet phones through paired appliances. You speak on the phone. Voider's screen and four buttons control setup.

## What you need

At each end: a Voider with its supported screen and buttons, a prepared microSD card, a Grandstream GXP1610 in factory settings, suitable power supplies and Ethernet cables.

At different places, each Voider needs wired Internet through a router. Paired Voiders also work on the same local network without Internet.

For pairing: one erasable USB stick. Keep each Voider's check code separately. This is not a package contents list.

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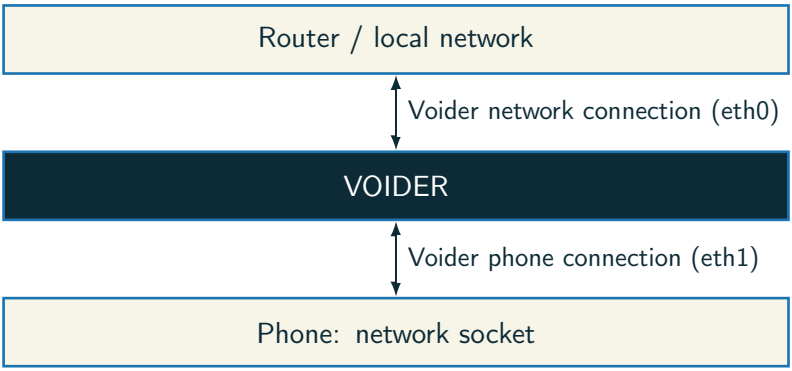
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**Review edition** Updated source, 11 October 2026. Read the open issues on page 15 before use.

# Connect the cables




Functional diagram. It does not show socket positions.

1. Connect cables with power disconnected.
2. Connect the phone's network socket to Voider's phone connection, not to the router.
3. Connect Voider's network connection to a router LAN socket.
4. Connect the intended power supplies to Voider and phone. Repeat at the other end.

You should see setup or **Device check**. After startup checks, **Net** shows **Phone**: **Connected** when the phone is reachable. A call still needs testing.

**Confirm hardware first** Enclosure labels, socket positions, and power ratings are not documented. Have them confirmed before connecting. Do not assume Ethernet supplies power.

# Buttons and language

The four buttons match the labels below them, left to right. Empty fields do nothing. Press and release for a normal action. Hold about one second, then release for a held action, marked by two white bars. A pressed label lights up immediately.



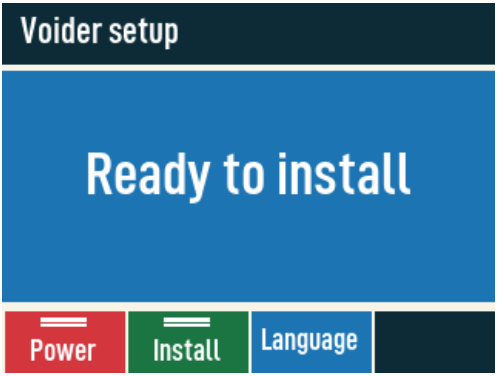

Real display rendering with example data.

1. From Home, press **Net**, then **Settings**, then **Language**.
2. Press **Next** until English appears. Press **Try**. The labels change now, for this running session.
3. Hold **Save** to keep the language after switching off. Record and confirm the new check code (page 5).

During first setup, press **Language**, then **Next**, **Try** and **Back**. There is no Save yet. Save after the startup checks. The starting language is English.

# First installation

Do this only if the screen shows **Voider setup**. An already installed Voider starts with **Device check**; continue on the next page.




The installation screen. Hold the second button to start.

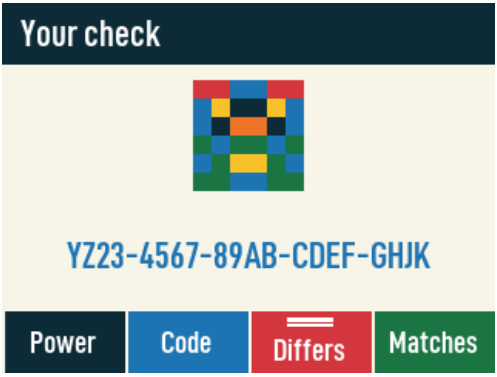
1. Select your language if needed (page 3). Hold **Install**.
2. Wait while the phase and percentage change. Keep power connected. The installer is already on the card; it does not need an Internet download.
3. At **Setup complete**, write down the displayed check code. Label your note for this particular Voider. It is your first reference.
4. Press **Restart** once. The display goes black during the restart. Wait for **Device check**.

Installation creates a unique identity. Repeat on the other Voider; keep its code separately. The restart needs your button press.

If **Setup failed** appears, note the message. Hold **Retry** after the cause is resolved, or hold **Power** to switch off.

# Check at every start

1. Wait for **Device check** to show **Passed**. This checks the installed system. Press **Next**.
2. At **Your check**, compare the code with your separate record for *this* Voider. **Code** shows the full fingerprint; **Back** returns.
3. If it matches, press **Matches**. You should see **Passed**. Press **Home**. Normal connections can now start.




Example only. Never use the code in this picture as your reference.

**Different or missing record?** Do not guess. If the code differs unexpectedly, hold **Differs**, then hold **Power**. Have the cause checked. A failed device check also blocks normal operation.

**After your own saved change: New check code** shows a new code. Write it down, check your copy, then press **I recorded it**. The previous screen returns directly. Use **Matches**/**Differs** only when comparing with your saved record at startup. Pairing, saved settings and deletion can change it. Keep the old note with the reason for the change. A normal clean restart does not change it.

# Pair: first Voider

Pairing gives two Voiders the information they need to connect. Choose one as A and the other as B. These are names for this guide, not screen labels. One transfer allows calls in both directions.




A: **Share** → Carry the stick → B: **Add**

1. Insert one USB storage stick and remove other storage sticks. On A, press **Links**, then **Share**. Voider selects the stick immediately. If it is missing, insert it and press **Retry**.
2. At **Erase USB?**, check the shown USB model. Hold **Erase** only if all files on that stick may be lost. **Cancel** leaves this screen without erasing.
3. Wait for **New check code**. Record and confirm A's new code as on page 5.
4. At **Safe to remove**, remove the stick and press **Done**. Carry the stick directly to B.

**Keep this stick private** It contains secret connection information and is not encrypted by Voider. Do not lend it to anyone else. Use a new Share operation for each different relationship.

# Pair: second Voider

1. Insert the stick from A. On B, press **Links**, then **Add**. Voider selects the stick immediately. If it is missing, insert it and press **Retry**. A confirmation screen shows **Key verified**. This screen does not show a pairing fingerprint. The bundle has been read and validated without changing the stick.
2. Hold **Add** again to import. This reads the stick; it does not erase it. Wait for **New check code**.
3. Record and confirm B's new check code. At **Safe to remove**, remove the stick and press **Done**. Store it safely.
4. Keep both Voiders on. Open the connection and wait for **Connected**. Then try a call.

**Two different checks** A's and B's startup check codes belong to different devices. They need not match each other. Compare each only with its own record.

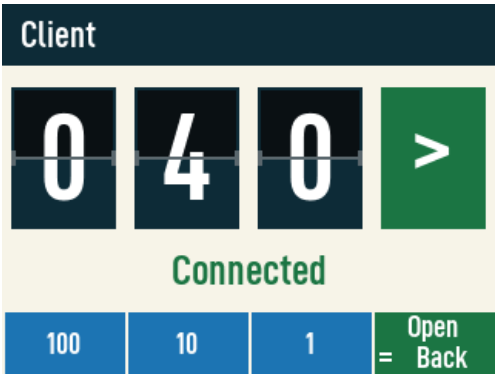
## Pairing fingerprint: current limit

The software records a separate fingerprint for the USB bundle. This candidate has no display page for comparing that fingerprint between A and B. **Key verified** is not a human identity check. If you require that comparison, stop here and have the product gap resolved. Do not substitute the startup codes.

If an error appears, follow page 12. Do not repeat Share just because the connection is still waiting.

# Go straight to a number

1. From Home, press **Links**, then **Open**. Choose **Clients** or **Servers**.
2. Three cards show the connection number. The first three buttons advance the hundreds, tens and units separately. The left card cycles 0–2; the other two cycle 0–9.
3. For Client 40, set **040**. From 001: leave the first card at 0, press the second button four times and the third nine times. Digits wrap to zero.
4. Press the fourth button, **Open**. Hold it to go back instead. An unsaved number shows **Not saved**; opening is disabled.

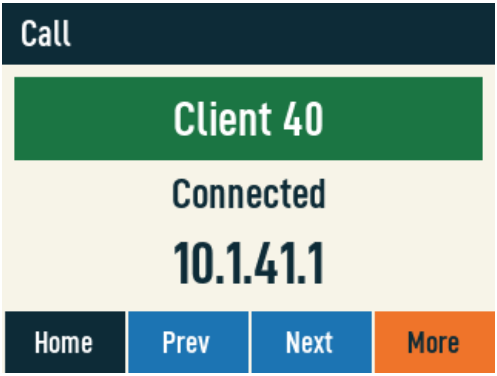



Example: Client 40 is saved and ready to open.

**Prev**/**Next** browse nearby entries. For another number: **More** → **Find**. Keep a note of each person's role and number: Client 40 and Server 40 differ.

# Call with the GXP1610

Open the saved connection as on the previous page. Wait for **Connected** and read the large address below **Call**.




Example: Client 40. Dial your own displayed address.

1. On the idle Grandstream GXP1610, press **MENU**. Use the arrows to select **Direct IP Call**, then press **MENU** again.
2. Enter the displayed address, using ***** for each dot. For 10.1.41.1 enter **10*1*41*1**. No port suffix is needed.
3. Use **More** to check that **IPv4** is selected. Press the **OK** softkey to dial.
4. When the other phone rings, lift its handset. Check sound both ways; hang up to end.

Either person can call. Voider's buttons do not dial. For a return call, use the Voider address, not a changing incoming 172.16.x.x caller address. No phone web login or configuration is needed.

# Manage relationships

**Add another person:** repeat pages 6–7 with a new **Share** on the chosen A. The existing list still offers **Share** and **Add**. You do not need to pair twice for two-way calls.

**Client** and **Server** describe local roles, not people. A Share creates a Client entry on A; Add creates a Server entry on B. Keep your own note of who uses each entry. The display does not offer renaming.

**Delete one relationship:**

1. Open the number as on page 8; check the entry name.
2. **More** → **Remove**. Check the entry name, then hold **Remove**.
3. Record and confirm the new code. The entry disappears locally. The other Voider's list is not automatically cleaned up. This device rejects that removed USB bundle on import.

**Retry one relationship:** Open the entry, choose **More** → **Retry**, then hold **Retry**. Pairing is kept; other connections keep running.

**Factory reset: Net** → **Settings** → **More** → **Reset**, then **Next** and held **Reset**. This removes all relationships and the SSH access key, switches SSH off and creates new device identities. Old peers need new pairing. Some settings, including language, remain. Record the new code. Do not use reset to fix a slow connection.

# Read the connection status

**Net** opens Internet and phone status. **Phone**: **Connected** means Voider can reach the phone. **Home totals:** 1/2 means one of two saved relationships is connected. Clients and Servers are counted separately. It is not a count of calls.

**Connected:** a path to the other Voider is active. **Connecting:** it is trying. **Offline:** no active path is shown. This alone does not prove that the other Voider is switched off.

Voider normally tries **LAN**, then **Direct**, then **Private** (WireGuard through routers), then **Tor**. The selected method appears under **More**. A healthy connection keeps its method until it fails, or you remove or disable it. LAN also uses WireGuard. Connections show **LAN4**, **LAN6**, **Direct4** or **Direct6** for the selected IP version.

**Give it time.** Clock checks and Tor startup can take minutes. A Tor recovery attempt has 90 seconds; the whole connection process can take longer. Leave both devices on. Re-pairing does not speed this up.

A same-network LAN connection can work without Internet. Phone readiness is checked separately.

**Paths:** normally leave all seven paths on. For an intentional change: **Up**/**Down** selects, **Remove**/**Add** switches availability. **Back** then held **Apply** saves; confirm the new code. **Cancel** returns to editing. At least one path must remain on. Restricting paths can prevent calls.

# If something does not work

1. Complete both startup checks on both Voiders.
2. Press **Net**. Check the phone's power and cable if missing; check the uplink cable and router if the network is missing.
3. Leave both devices on for several minutes. If still offline: **Network** → **Retry**, then hold **Retry**. This may interrupt a call.
4. If it still fails, note the screen message and connection number. Keep pairing and check-code records. Do not routinely erase or reset.

**USB errors:** Use one stick. Mounted or unsafe media are refused. After an error, reopen the action and select the stick again.

**No sound or ringing:** recheck the address and both phones. A green connection or phone ping does not prove speech. Phone web administration is unnecessary.

**Switch off:** Home → **Power**, then hold **Power**. Wait for shutdown before disconnecting power; a black panel alone does not prove completion.

**Restart:** Home → **Power** → **Restart**, then hold **Restart**. Pairing stays saved. Complete the normal checks after boot.

# Save a backup

A backup contains this device's saved identity, relationships, keys and settings. It does not back up the phone or the whole system card. The files are not encrypted. Keep the stick as carefully as a key to your device.

1. From Home: **Net** → **Settings** → **Backup**.
2. Insert one spare USB stick. Hold **Backup**.
3. Read **Erase USB?**. All files on that stick will be erased. Hold **Erase** only if that is intended.
4. Wait for **Safe to remove**, then remove the stick. Label it with the device, date and its current check code. Keep separate backups for A and B.

## Restore

Insert the backup stick. Open **Net** → **Settings** → **More** → **Restore**, then press **Check**. Voider reads and validates the backup without changing the stick. It must match the installed system.

Hold **Restore** only to replace all saved settings, identities and connections with this backup. Later changes are lost. Record the new check code and press **I recorded it**. Remove the stick, then hold **Restart** to apply. Old running settings cannot overwrite the restored backup. After restart, compare the recorded code using **Matches** or **Differs**.

# Optional: remote access

SSH lets a trusted administrator manage Voider from a computer. It is not needed for calls. Open **Net** → **Settings** → **More** → **SSH**. The screen shows **SSH on** or **SSH off**.

**Switch on or off:** press **On** or **Off**, then hold the same label on the confirmation screen. Record and confirm the new code. Off blocks new management logins but keeps the key. Existing sessions may remain. The internal call transport is separate.

**Create or replace a key:**

1. Select **Login key**. Read the create or replace question. Replacement makes the old login key stop working. Hold **OK**.
2. Insert one erasable stick and press **Check**. Hold **Erase** to erase all its partitions and files. Voider writes and verifies the new key before replacing access.
3. Record and confirm the new code. SSH is now on. At **Safe to remove**, remove the stick and press **Done**. Protect the stick. It holds `voider-admin` (private login key) and `voider-admin.pub` (public key). The private key has no passphrase.

**Remove permanently:** select **Remove**, then hold **Remove**. This revokes the login key and leaves SSH off. Record the new code. A USB failure keeps the previous key and access setting. Voider keeps only the public half; it cannot recover a lost private key.

# What the checks mean

**Device check** checks the installed system against its recorded values. **Your check** lets you compare saved state with your own separate record. Neither proves the identity of the person holding the other phone.

**Pairing** establishes shared secrets. The current display cannot show the pairing fingerprint for human comparison (page 7). Protect pairing sticks and backups.

**Network protection:** WireGuard protects the LAN, direct and router-traversal paths. On the Tor path, Tor supplies encryption. The tundup component supplies authenticated plaintext: it checks authenticity and rejects replays, but does not encrypt the call data itself. Do not assume the local phone cable is encrypted.

The custom protocol has not been independently audited. Voider does not guarantee anonymity or prevent traffic analysis. Stolen pairing secrets can also put past sessions at risk.

**SSH boundary:** root access needs a key and a private network on the Ethernet uplink. No passwords, phone-side, public-WAN or Tor login, or forwarding. A prepared key enables installation SSH; no key means off. Installed devices require both startup checks. The administrator must verify the host key. The display has no host-key page or factory SSH switch.

**Before customer release:** confirm ports, power and factory SSH controls; resolve pairing verification. Native language and beginner review remain needed. Fresh installations and physical acceptance remain open. The Tor observation was 601 seconds, not one hour. This manual does not declare the image ready for sale.